

## FORAGE CROPS

This topic entails the following:

- Definition and classification of pasture crops.
- Identification of pasture crops
- Description of ecological requirements of forage crops
- Description of the establishment of pasture and fodder crops
- Description of forage utilization and conservation.

The following relevant questions and their answers in this topic will greatly motivate and help the user to comprehend and understand the required concepts and practices.

1. State **three** factors which affects the quality of standing forage given to livestock
2. Explain Napier grass production under the following sub-headings
  - i) Seedbed preparation
  - ii) Planting
  - iii) Fertilizer application
  - iv) Weed control
  - v) Utilization
3. Why is it necessary to allow freshly cut nappier grass to wilt before ensiling
4. Describe the establishment of grass pasture from the time the land is ploughed using a mould board plough to the time the pasture is ready for grazing
5. State **two** causes of failure in pasture establishment
6. State **two** advantages of grass-legume mixture
7. List **four** factors that determine the quality of hay
8. i) Discuss the production of Guatemala grass (*Trypsacum Laxum*) under the following headings
  - a) Ecological required
  - b) Land preparation
  - c) Planting
  - d) (i) Utilization and defoliation
- ii) Discuss **six** effects of late defoliation of fodder
9. Give **four** factors that determine the nutrient content in hay
10. (a) Describe the preparation of the following farm materials:-
  - (i) Farm yard manure
  - (ii) Hay
- (b) Explain the factors considered in timely planting of annual crops
11. (a) What is topping in pasture management  
(b) State two methods used in topping in pasture management
12. Give **two** advantages of grass-legume pasture over pure grass pasture
13. Why are farmers encouraged to conserve excess forage in the farm?
14. Give **two** factors affecting the quality of hay

## FORAGE CROPS

1. three factors which affects the quality of standing forage given to livestock
  - Forage species
  - Stage of harvesting
  - Mode of feeding
2. i) Seedbed preparation
  - Done during dry period/ done early
  - Clear vegetation/ remove stumps
  - Carry primary cultivation/ harrowing to
  - Make furrows/ holes
  - Spacing 90cm x 50cm for cutting and 90cm x 50cm for splits

---

## ii) Planting

- At on set of rains/ early planting/ irrigate if necessary
- Select variety for ecological condition of the area
- Use health planting material
- Place the planting material in holes/ furrows
- Cover the planting material with soil to an appropriate depth
- Use cuttings or splits
- Select cutting from mature cane/ stems
- Cutting should have 3-5 nodes

---

## iii) Fertilizer application

- Apply phosphatic fertilizer at planting 200kg/ha
- Apply manure before planting, 7-10 tonnes/ha
- Top dress phosphorous fertilizer after 8 weeks after planting
- Apply manure after harvesting and dig it

---

## iv) Weed control

- Cultivation/ tillage/ mechanical
- Uprooting
- Slashing
- Suitable herbicide application
- Interplanting legumes as cover crops e.g. Desmodium.

---

## v) Utilization

- Cut and take to animals when proportion of leaf is higher than the stem/ 3-5 months after planting/ frequency 8 weeks
- Cut down excess foliage to conserve as silage or hay
- Cut and sold
- Cut when mature to get stem cuttings for planting
- Cut stems at 2.5 – 5cm above the ground surface
- Use a sharp panga for harvesting
- Chop forage into small pieces before feeding
- Na[pier grass is cut, dried and used as mulching material

3. to reduce moisture content ✓ which can lead to rotting instead of formation during ensiling

4. (a) - Harrow the land to a fine filth;

- Harrow during the dry or before the rains;
- Make the seed be weed – free / ensure clean seed bed;
- Firm the seed bed using rollers after sowing;
- Select a desirable variety of seed for the ecological zone.;
- Sow seeds at the onset rains/ early planting;
- Apply phosphatic fertilizers at appropriate rate of 200 – 300 kgs/ ha at planting time;
- Drill or broadcast the seeds evenly;
- Use a recommended seed rate for the variety / seed rate of 1.5 – 2.0 kh/ha pure seeds;
- Bury seeds at 2 ½ times their diameter;
- Control weeds by uprooting/ apply a suitable herbicide;
- Apply nitrogenous fertilizers about 6 weeks after germination in split application.
- Avoid grazing when the pasture is too young.
- Practice light grazing in the field phase of pasture establishment. (10 x 1 = 10 mk)

5. Two causes of failure in pasture establishment

- Poor seed germination due to wrong placement of seeds
- Poor inoculation of legume seeds
- Lack f nutrients in the soil
- Unfavourable chemical conditions in the soil
- Poor drainage
- Pest and disease attacks

14. - High nutritive value

- relieve bloat
- Higher yields of forage per unit area
- Improve soil fertility due to nitrogen fixation
- Economy in use of nitrogen fertilizers

6. four factors that determine the quality of hay

- Forage species used
- Stage of harvesting i.e. leaf-stem ratio
- Length of drying period
- Weather condition during process
- Conditions of storage structure

7. a) Altitude - High altitude  
- 2000 m above sea level

Soils - well drained  
- deep fertile

Rainfall High rainfall above 900mm per annum  
Well distributed throughout the growing period

b)

- Prepare land early enough/ during dry season/ before onset of rains
- Carry out primary cultivation appropriately
- Harrow to a medium tilth
- Remove all perennial weeds

c) - Establishment from stem cutting or splits or seeds makes furrows at a spacing 1m apart. Plant the grass 0.5m apart within the rows. Holes may also be used  
Planting is done on the onset of long rains  
Phosphate fertilizers are used as planting fertilizers  
DAP fertilizer is applied at the rate of 100 – 150kg/ ha  
Organic manure is applied at the rate 10 tones/ ha

d)

Utilization – chopped and fed to livestock as green fodder

Detoliation – can be harvested when it is over 8 – 12 weeks

ii)

- Forage has high DM content hence high DM yield
- High cellulose content hence it is woody and fibrous
- High lignin cuten lannin and silia content which are all indigestible
- It has low crude protein content
- It has low leaf stem – ratio
- It has low dry matter digestibility

8. – Weather conditions during dry process;

-Length of the drying period

- Stage of growth at harvesting time/leaf-stem ratio of the plant species;

- Species of the hay crop;

- Storage facilities/method of storage;

- Period of storage

- Disease and pest attack on the crop;

- Fertility of the soil;

9. (a) (i) Preparation of farm yard manure:-

- Collect animal waste/refuse/dung and urine;
- Collect animal bedding/litter and other rotten plant residues;
- Store collected materials under roof/shed to prevent leaching and oxidization of nutrients;
- Turnover the materials regularly;
- Sprinkle water if dry;
- leave the material to rote completely before use; (6x1=6mks)

(ii) Preparation of Hay

- Cut the grass /legume in the field when 50% of it is starting to flower;
- The cut forage is spread in the field for four continuous days (sunny days)
- The cut forage is turned daily for even for four uniform drying;
- Gather the dried material in a central spot;
- Bale the material;
- Properly store the baled hay (6x1=6mks)

(b) Factors to consider in timely planting of annual crops

- Escape from serious weed competition;
  - Utilization of early rainfall;
  - Exploitation of Nitrogen flush in the soil that has accumulated during dry season;
  - Escape from serious pest + disease attack e.g. stalk borer in maize;
  - Fetch high market prices when harvested early;
  - Reduce competition for labour during labour peak period;
  - For harvesting season to coincide with dry period to reduce losses e.g. cotton
- Early planting means early farming/calendar for the farmer to enable him /her to finish up other farm activities; (8x1=8mks)

10. (a) It is the constant removal of the sterony fibrous material left behind after continuous grazing

- (b) -Done by slashing the whole pasture to the base, leaving only the maintenance forage.
- Burning is also done.

11. two advantages of grass-legume pasture over pure grass pasture

- More nutritious to livestock
  - Improves soil fertility through nitrogen fixation
  - More total yield per unit area
  - Security against total pasture lose (2x1=2mks)
- to feed animals during the dry seasons
- forage species used
- Stage of harvesting / leaf to stem ratio
- Length of drying period
- Weather conditions
- Storage conditions

### **INTERVIEW GRID (completion)**

Evaluation of the MOOC on knowledge translation

~~Before we begin the interview and recording, I'd like to quickly introduce myself. My name is RV. I'm doing a master's degree in psychology at the Université de Montréal under the supervision of Professor CD. My thesis project involves evaluating the effectiveness and quality of the MOOCs on KT developed by the RENARD research team. Thus, your participation is solicited to evaluate the content, structure, progress, and potential effects of the MOOC on your practices.~~

~~I'm going to ask you around forty questions, divided into 4 sections.~~

~~The recording will be kept securely by me, and only members of the research team will have access to its contents.~~

~~Do you agree to the interview being recorded?~~

~~During the interview period, don't hesitate to expand on your answers, even if it's outside the scope of the question.~~

~~The interview is likely to last between 45 and 60 minutes.~~

***Do you have any questions before we get started?***

***I'm going to start with some questions that will allow me to get to know you and the context in which you've taken the MOOCs.***

~~1. Where did you access the course (country)?~~

~~2. What was the last diploma you obtained?~~

~~3. What job/position do you currently hold?~~

~~4. Could you briefly describe your organization and its missions?~~

~~5. How important is knowledge translation in your work? Why or why not?~~

~~● Does this have any bearing on your professional activities?~~

~~● Do you think that developing your KT skills is useful for you? In what way?~~

~~6. For those who have requested a certificate of completion:~~

~~● How did you use this certificate?~~

~~7. Why did you take this course? What were your motivations?~~

~~8. A slightly different question now, what do you think helped you make it to the end of the course?~~

**~~Now I'm going to ask you some related to the technology and accessibility of the course.~~**

---

#### **Technological factors/accessibility**

---

9. ~~What did you use to access the course? (Laptop, tablet...)~~
- ~~● Would you prefer to have access via your smartphone?~~
  - ~~● Was the content compatible with the IT tool you used? For example, was your computer or its system recent enough to host the platform?~~
10. ~~Did you encounter any difficulties connecting? Or any other technical/technological difficulties?~~
- ~~● Do you have a good Internet connection?~~
  - ~~● Has this affected your experience?~~
11. ~~Did you find the course interface user friendly? What would you change (font size, colors, video/image quality)?~~
- ~~● Was the course structure well organized and easy to follow? If not, what would you change?~~
12. ~~Did you use the discussion forum? Was it relevant? Why or why not?~~

**~~I'm going to ask you some questions that focus on the content of the course.~~**

---

#### **Pedagogical factors and content assessment**

---

13. ~~Based on your lifestyle/values/culture, your job, does the course lack personalization/adaptability of content? Did it offend your values?~~
14. ~~Do the subjects correspond closely enough to your reality?~~
- ~~● Were there any examples you didn't like? Which ones?~~
  - ~~● Is the language appropriate?~~
15. ~~What did you think of the exams? Were they too difficult?~~
16. ~~Was the general content relevant? What would you add?~~
17. ~~Do you plan to refer to and re-use the notes, resources and summary sheets provided during the course?~~
- ~~● Do you still use the materials provided (summary sheets, readings) or do you revisit the platform if necessary?~~
18. ~~Was there enough variety in the materials provided? Quality?~~

- ~~Which activity(ies) did you prefer (educational videos, quizzes, readings, practical exercises, Vox pop, summary sheets)? Why?~~

- ~~Which activity(ies) did you enjoy least? Why?~~

19. ~~What do you think of the references/resources/readings available to you (quantity/quality)?~~

- ~~Have you read them? (All, most, just a few...)~~

20. ~~What do you think would be the best way to get professional learners involved in such activities?~~

**~~Now I'm going to ask you some knowledge-based questions. We've reached the 20<sup>th</sup> question (half).~~**

---

#### **~~Individual factors~~**

---

21. ~~Did you already have basic knowledge of the course notions (in KT)?~~

22. ~~Do you feel you lack computer skills? For example, are you able to solve problems or perform basic tasks?~~

23. ~~Do you enjoy taking online courses? Why or why not?~~

24. ~~Did you find the course easy to use?~~

25. ~~Was the asynchronous mode relevant? Or did you feel a lack of interaction?~~

- ~~With an instructor~~
- ~~With other learners~~

26. ~~Do you find this MOOC type course format attractive? Why or why not?~~

27. ~~In your opinion, what does an online course like this offer (or not) compared to a face-to-face course?~~

- ~~If you were to make a comparison between the two (face-to-face versus distance learning), what would be the advantages of the MOOC?~~
- ~~What are the disadvantages of the MOOC?~~

28. ~~Has taking this course put any constraints on your schedule? How did you organize your time (work/family/study)?~~

- ~~During the e-learning sessions, were you in a supportive, distraction-free environment?~~
- ~~Did you have the support of your family and friends or your employer?~~

29. ~~What are your expectations for a future course like this?~~

**~~I'll move on to questions about the potential impact the course has had on your practices.~~**

---

#### **~~Application of knowledge and potential impact of the MOOC~~**

---

30. ~~Do you have or plan to apply any of what you've learned in the course? Which ones?~~

- ~~How can I help? What do you think will be most useful to you? Why or why not?~~

31. How do you put what you've learned to practical use? Could you give me an example of a KT application?

32. Have any of your practices changed?

- Can you give me a concrete example of a change in your practices?
- How has this changed the way you do business?

33. Has anything changed in the way you communicate?

- Can you give me a concrete example related to your practice?

34. What are the barriers to changing KT practices?

35. Are the people around you (researchers, people you work with) open to KT?

36. Would you recommend this course to others? Why or why not?

---

#### Other

---

37. Do you have any other comments? Is there anything else you'd like to talk about?

- What do you think could be done to improve this course?

— Thank you for your cooperation!